

Cramér–Rao-type bounds for quantum multiparameter estimation with separable measurements

Mankei Tsang^{1,2,*}

¹*Department of Electrical and Computer Engineering,
National University of Singapore, 4 Engineering Drive 3, Singapore 117583*

²*Department of Physics, National University of Singapore, 2 Science Drive 3, Singapore 117551*
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This memo fills the gaps in the proofs of some Cramér–Rao-type bounds for quantum multiparameter estimation with separable measurements, such as the Nagaoka–Hayashi bound.

1. INTRODUCTION

There has been a lot of recent interest in Cramér–Rao-type bounds for quantum multiparameter estimation with separable measurements. The so-called Nagaoka–Hayashi bound [1–3] is one interesting example, but a proof of the bound for N independent and identically distributed (IID) objects and separable measurements does not seem to be available in one place. The ingredients are available in Refs. [4, 5], but the arguments there are not obvious enough for the author’s taste. This memo fills the gap.

2. MOST INFORMATIVE CRAMÉR–RAO BOUND

Define a quantum statistical model as

$$\{\rho(\theta) : \theta \in \Theta\}, \quad (2.1)$$

where θ is an unknown parameter and each $\rho(\theta)$ is a density operator on a complex Hilbert space $\mathcal{H}$. Let $\beta : \Theta \rightarrow \mathbb{R}^q$ be a vectoral parameter of interest. Define the error matrix

$$V_{jk}(\rho(\theta), E, \check{\beta}) \equiv \int [\check{\beta}_j(\omega) - \beta_j(\theta)] [\check{\beta}_k(\omega) - \beta_k(\theta)] P_{\rho(\theta), E}(d\omega), \quad (2.2)$$

$$P_{\rho(\theta), E}(\cdot) \equiv \text{tr } E(\cdot) \rho(\theta), \quad (2.3)$$

where E is a POVM on $\mathcal{H}$ with sample space Ω and $\check{\beta} : \Omega \rightarrow \mathbb{R}^q$ is an estimator. Most prior work assumes $\beta(\theta) = \theta$, but it is easy to account for an arbitrary $\beta(\theta)$. Assume that everything is finite-dimensional. In particular, let the parameter space $\Theta \subseteq \mathbb{R}^p$ be Euclidean and p -dimensional. Assume also that everything is well defined, e.g., all functions are as differentiable as needed and all matrices are invertible when needed.

We will need to consider only a neighborhood around one θ called the true parameter; all functions of θ will be evaluated at the true θ . Define any function $\eta(\cdot, \theta) : \Omega \rightarrow \mathbb{R}^q$ at the true θ that satisfies

$$\int \eta_j(\omega, \theta) P_\theta(d\omega) = 0, \quad \int \eta_k(\omega, \theta) \partial_j P_\theta(d\omega) = \partial_j \beta_k, \quad \partial_j \equiv \frac{\partial}{\partial \theta_j} \quad (2.4)$$

as a so-called influence function [6]. Denote the set of all influence functions as $\mathcal{D}(P_\theta)$. If $\check{\beta}$ is locally unbiased at the true θ , Then $\check{\beta}(\omega) - \beta(\theta) \in \mathcal{D}(P_\theta)$. It must be stressed that, given an influence function $\eta(\omega, \theta)$ for all $\theta \in \Theta$, there may not exist a θ -independent estimator such that $\check{\beta}(\omega) = \beta(\theta) + \eta(\omega, \theta)$. Write

$$\Sigma_{jk}(\rho(\theta), E, \eta) \equiv \int \eta_j(\omega, \theta) \eta_k(\omega, \theta) P_{\rho(\theta), E}(d\omega). \quad (2.5)$$

* mankei@nus.edu.sg; <https://blog.nus.edu.sg/mankei/>

Abbreviate the true $\rho(\theta)$ as ρ . Let W be a $q \times q$ real, symmetric, positive-semidefinite matrix and consider the so-called most informative Cramér–Rao bound (CRB) [1, Sec. 5]

$$\text{MI}(\rho) \equiv \inf \{ \text{tr } W \Sigma(\rho, E, \eta) : E \in \{\text{POVM}\}, \eta \in \mathcal{D}(P_{\rho, E}) \} \quad (2.6)$$

$$= \inf \{ \text{tr } W \Sigma(\rho, E) : E \in \{\text{POVM}\} \}, \quad (2.7)$$

$$\Sigma(\rho, E) \equiv (\partial \beta)^\top J(\rho, E)^{-1} \partial \beta, \quad (\partial \beta)_{jk} \equiv \partial_j \beta_k, \quad (2.8)$$

where $J(\rho, E)$ is the Fisher information matrix and $\Sigma(\rho, E)$ is the matrix CRB for the classical statistical model

$$\{P_{\rho(\theta), E}(\cdot) : \theta \in \Theta\}. \quad (2.9)$$

MI is also called the quasiquantum CRB [4, Eqs. (4) and (5)]. The proof of Eq. (2.7) can be found, for example, in Ref. [6]. Assume for simplicity that the sample space is countable. Define the score functions

$$S_j(\omega, \theta) \equiv \partial_j \ln P_{\rho(\theta), E}(\omega), \quad j = 1, \dots, p. \quad (2.10)$$

The Fisher information becomes

$$J_{jk}(\rho, E) \equiv \sum_{\omega} P_{\rho, E}(\omega) S_j(\omega, \theta) S_k(\omega, \theta). \quad (2.11)$$

The $\eta \in \mathcal{D}(P_{\rho, E})$ that achieves the infimum in Eq. (2.6) for a given E is called the efficient influence function, given by

$$\eta_j^{\text{eff}}(\omega, \theta) = \sum_{k, l} (\partial_k \beta_j) [J(\rho, E)^{-1}]_{kl} S_l(\omega, \theta). \quad (2.12)$$

A separable POVM E on $\mathcal{H}^{\otimes N}$ with sample space $\Omega^{(N)}$ is by definition in the form

$$E(\omega) = \sum_r E_1(\omega, r) \otimes E_2(\omega, r) \otimes \dots \otimes E_N(\omega, r), \quad \omega \in \Omega^{(N)}, \quad \sum_{\omega} E(\omega) = I^{\otimes N}. \quad (2.13)$$

An important special case is a product POVM in the form

$$E(\omega) = E_1(\omega) \otimes E_2(\omega) \otimes \dots \otimes E_N(\omega). \quad (2.14)$$

Given any separable POVM in the form of Eq. (2.13), we can construct a product POVM as

$$E'(\omega, r) = E_1(\omega, r) \otimes E_2(\omega, r) \otimes \dots \otimes E_N(\omega, r), \quad (2.15)$$

such that, by monotonicity, $J(\rho, E) \leq J(\rho, E')$. This fact allows us to consider only product POVMs in what follows.

Define the IID model as

$$\{\rho(\theta)^{\otimes N} : \theta \in \Theta\}. \quad (2.16)$$

The following lemma and theorem are essentially the solution to [5, Exercise 6.42].

Lemma 1. *For any product POVM E on $\mathcal{H}^{\otimes N}$, there exists a POVM $\tilde{E}$ on $\mathcal{H}$ such that the Fisher information $J(\rho^{\otimes N}, E)$ for the IID model satisfies*

$$\frac{1}{N} J(\rho^{\otimes N}, E) = J(\rho, \tilde{E}), \quad (2.17)$$

where $J(\rho, \tilde{E})$ is the single-copy Fisher information at ρ .

Proof.

$$J_{jk}(\rho^{\otimes N}, E) = \sum_{\omega} \left[\prod_l \text{tr } E_l(\omega) \rho \right] \left[\partial_j \ln \prod_n \text{tr } E_n(\omega) \rho(\theta) \right] \left[\partial_k \ln \prod_m \text{tr } E_m(\omega) \rho(\theta) \right] \quad (2.18)$$

$$= \sum_{n,m} K_{jk}(n, m), \quad (2.19)$$

$$K_{jk}(n, m) \equiv \sum_{\omega} \left[\prod_l \text{tr } E_l(\omega) \rho \right] [\partial_j \ln \text{tr } E_n(\omega) \rho(\theta)] [\partial_k \ln \text{tr } E_m(\omega) \rho(\theta)]. \quad (2.20)$$

If $n \neq m$,

$$K_{jk}(n, m) = \sum_{\omega} \left[\prod_{l \neq n, m} \text{tr } E_l(\omega) \rho \right] [\partial_j \text{tr } E_n(\omega) \rho(\theta)] [\partial_k \text{tr } E_m(\omega) \rho(\theta)] \quad (2.21)$$

$$= \sum_{\omega} \text{tr} [E_1(\omega) \otimes \cdots \otimes E_N(\omega)] [\dots \rho \otimes \partial_j \rho(\theta) \otimes \rho \dots \rho \otimes \partial_k \rho(\theta) \otimes \rho \dots] = 0. \quad (2.22)$$

This leaves

$$J(\rho^{\otimes N}, E) = \sum_n K(n, n). \quad (2.23)$$

The next trick is to define

$$\tilde{E}_n(\omega) \equiv E_n(\omega) \prod_{l \neq n} \text{tr } E_l(\omega) \rho. \quad (2.24)$$

Note that ρ here is the true ρ ; it's assumed to be fixed and not to vary with θ for the purpose of the proof. Note also

$$\sum_{\omega} \tilde{E}_n(\omega) = \sum_{\omega} E_n(\omega) \prod_{l \neq n} \text{tr } E_l(\omega) \rho = \sum_{\omega} \text{tr}_{\neq n} [E_1(\omega) \otimes \cdots \otimes E_N(\omega)] [\rho \otimes \dots I \cdots \otimes \rho] \quad (2.25)$$

$$= \text{tr}_{\neq n} (\rho \otimes \dots I \cdots \otimes \rho) = I, \quad (2.26)$$

so $\tilde{E}_n$ is a legal POVM on $\mathcal{H}$. We can then write

$$K_{jk}(n, n) = \sum_{\omega} [\text{tr } \tilde{E}_n(\omega) \rho] [\partial_j \ln \text{tr } \tilde{E}_n(\omega) \rho(\theta)] [\partial_k \ln \text{tr } \tilde{E}_n(\omega) \rho(\theta)] = J_{jk}(\rho, \tilde{E}_n), \quad (2.27)$$

$$J(\rho^{\otimes N}, E) = \sum_n J(\rho, \tilde{E}_n). \quad (2.28)$$

Now define another POVM on $\mathcal{H}$ as

$$\tilde{E}(\omega, n) = \frac{1}{N} \tilde{E}_n(\omega), \quad \sum_{\omega} \sum_{n=1}^N \tilde{E}(\omega, n) = \sum_{\omega} \sum_{n=1}^N \frac{1}{N} \tilde{E}_n(\omega) = I. \quad (2.29)$$

$\tilde{E}$ is implemented by choosing a POVM randomly from $\{\tilde{E}_1, \tilde{E}_2, \dots, \tilde{E}_N\}$ with equal probability $1/N$. The Fisher information of $\tilde{E}$ is

$$J(\rho, \tilde{E}) = \frac{1}{N} \sum_n J(\rho, \tilde{E}_n), \quad (2.30)$$

which together with Eq. (2.28) proves the lemma. $\square$

Theorem 1. Let $\Sigma(\rho^{\otimes N}, E)$ be the CRB for the IID model. Define

$$N\text{MI}_{\text{sep}}(\rho^{\otimes N}) \equiv \inf \{N \text{tr} W \Sigma(\rho^{\otimes N}, E) : E \in \{\text{separable POVM}\}\} \quad (2.31)$$

$$= \inf \{N \text{tr} W \Sigma(\rho^{\otimes N}, E) : E \in \{\text{product POVM}\}\}. \quad (2.32)$$

Then

$$\boxed{N\text{MI}_{\text{sep}}(\rho^{\otimes N}) = \text{MI}(\rho)}, \quad (2.33)$$

where $\text{MI}(\rho)$ is the single-copy most informative bound.

Proof. To prove Eq. (2.32), note that $\{\text{product POVM}\} \subseteq \{\text{separable POVM}\}$ and we can always construct a product POVM E' from a separable POVM E with $J(\rho, E) \leq J(\rho, E')$, so we can consider only product POVMs for the infimum. For any product POVM E , Lemma 1 gives

$$N\Sigma(\rho^{\otimes N}, E) = (\partial\beta)^\top [J(\rho^{\otimes N}, E)/N]^{-1} \partial\beta = (\partial\beta)^\top J(\rho, \tilde{E})^{-1} \partial\beta = \Sigma(\rho, \tilde{E}). \quad (2.34)$$

Hence

$$N\text{MI}_{\text{sep}}(\rho^{\otimes N}) = \inf_{\text{product } E} \text{tr} W \Sigma(\rho, \tilde{E}) \geq \text{MI}(\rho). \quad (2.35)$$

To prove that the last inequality is an equality, let F be an arbitrary POVM on $\mathcal{H}$ with sample space Ω and construct a product POVM $E^{(F)}$ on $\mathcal{H}^{\otimes N}$ by

$$\Omega^{(N)} = \Omega^N, \quad \omega = (\omega_1, \dots, \omega_N), \quad \omega_n \in \Omega, \quad E_n^{(F)}(\omega) = F(\omega_n), \quad (2.36)$$

so that $(\omega_1, \dots, \omega_N)$ are N IID outcomes. Then

$$J(\rho^{\otimes N}, E^{(F)}) = NJ(\rho, F), \quad (2.37)$$

$$\inf_{\text{product } E} \text{tr} W N \Sigma(\rho^{\otimes N}, E) \leq \inf_F \text{tr} W N \Sigma(\rho^{\otimes N}, E^{(F)}) = \inf_F \text{tr} W \Sigma(\rho, F) = \text{MI}(\rho). \quad (2.38)$$

□

To generalize the bound for asymptotically unbiased estimators, define the estimator mean as

$$\bar{\beta}(\rho(\theta)^{\otimes N}, E, \check{\beta}) \equiv \int \check{\beta}(\omega) \text{tr} E(d\omega) \rho(\theta)^{\otimes N}. \quad (2.39)$$

For any biased or unbiased estimator,

$$V(\rho^{\otimes N}, E, \check{\beta}) \geq (\partial\bar{\beta})^\top J(\rho^{\otimes N}, E)^{-1} \partial\bar{\beta}. \quad (2.40)$$

For an asymptotically unbiased estimator, $\bar{\beta} \rightarrow \beta$ in some neighborhood of the true θ as $N \rightarrow \infty$, so the single-copy MI still applies to those in the asymptotic limit [4].

3. NAGAOKA-HAYASHI BOUND

Let

$$L_{jk}(E, \eta) \equiv \sum_{\omega} \eta_j(\omega) \eta_k(\omega) E(\omega), \quad \delta_j(E, \eta) \equiv \sum_{\omega} \eta_j(\omega) E(\omega). \quad (3.1)$$

Regard L and $\delta\delta^\top$ as operators on the complex $\mathcal{H}^{\oplus q}$, where each element of $\mathcal{H}^{\oplus q}$ is written as

$$|\psi\rangle = \bigoplus_j |\psi_j\rangle, \quad (3.2)$$

such that

$$L|\psi\rangle = \bigoplus_j \sum_k L_{jk} |\psi_k\rangle, \quad \langle\phi|L|\psi\rangle = \sum_{j,k} \langle\phi_j|L_{jk}|\psi_k\rangle, \quad (3.3)$$

$$\delta\delta^\top|\psi\rangle = \bigoplus_j \delta_j \sum_k \delta_k |\psi_k\rangle, \quad \langle\phi|\delta\delta^\top|\psi\rangle = \sum_{j,k} \langle\phi_j|\delta_j\delta_k|\psi_k\rangle. \quad (3.4)$$

Lemma 2 ([3]). *For any POVM,*

$$L \geq \delta\delta^\top. \quad (3.5)$$

Proof. For any $|\psi\rangle \in \mathcal{H}^{\oplus q}$, write

$$\langle\psi|L|\psi\rangle = \sum_\omega \langle f_\omega|E(\omega)|f_\omega\rangle, \quad \langle\psi|\delta\delta^\top|\psi\rangle = \sum_{\omega,\tau} \langle f_\omega|E(\omega)E(\tau)|f_\tau\rangle, \quad (3.6)$$

$$|f_\omega\rangle \equiv \sum_j \eta_j(\omega) |\psi_j\rangle. \quad (3.7)$$

Then

$$\sum_\omega \left(\langle f_\omega| - \sum_\tau \langle f_\tau|E(\tau) \right) E(\omega) \left(|f_\omega\rangle - \sum_\tau E(\tau)|f_\tau\rangle \right) \geq 0, \quad (3.8)$$

$$\sum_\omega \langle f_\omega|E(\omega)|f_\omega\rangle - \sum_{\omega,\tau} \langle f_\omega|E(\omega)E(\tau)|f_\tau\rangle \geq 0, \quad (3.9)$$

$$\langle\psi|L|\psi\rangle - \langle\psi|\delta\delta^\top|\psi\rangle \geq 0. \quad (3.10)$$

□

Instead of minimizing over (E, η) for MI, we can obtain a lower bound on MI by minimizing over all (L, δ) that satisfy the constraints. The result is the Nagaoka–Hayashi bound (NH).

Theorem 2 (Nagaoka–Hayashi bound [3]).

$$\boxed{\text{MI}(\rho) \geq \text{NH}(\rho)}, \quad (3.11)$$

$$\text{NH}(\rho) \equiv \inf \left\{ \text{tr } W\Sigma : \Sigma_{jk} = \text{tr } L_{jk}\rho, L_{jk} \in \mathcal{O}(\mathcal{H}), L = L^\top, L \geq \delta\delta^\top, \delta \in \mathcal{D}(\rho) \right\}, \quad (3.12)$$

where $\mathcal{O}(\mathcal{H})$ is the set of self-adjoint operators on $\mathcal{H}$, $(L^\top)_{jk} \equiv L_{kj}$, and $\mathcal{D}(\rho)$ is the set of influence operators, defined as

$$\mathcal{D}(\rho) \equiv \left\{ (\delta_1 \ \dots \ \delta_q)^\top : \delta_j \in \mathcal{O}(\mathcal{H}), \text{tr } \delta_j\rho = 0, \text{tr } \delta_j\partial_k\rho(\theta) = \partial_k\beta_j \right\}. \quad (3.13)$$

Proof. $L(E, \eta)$ and $\delta(E, \eta)$ satisfy the constraints in the infimum for any POVM E and any influence function η . □

Theorems 1 and 2 together imply

$$\boxed{N\text{MI}_{\text{sep}}(\rho^{\otimes N}) = \text{MI}(\rho) \geq \text{NH}(\rho)}, \quad (3.14)$$

so the Nagaoka–Hayashi bound applies to the IID model with separable measurements.

4. HOLEVO–NAGAOKA BOUND

Now define the Holevo–Nagaoka bound, also called the Holevo Cramér–Rao bound, as [1, 7, 8]

$$\text{HN}(\rho) \equiv \inf \left\{ \text{tr } W \Sigma : \Sigma \in \mathbb{R}^{q \times q}, \Sigma = \Sigma^\top, \Sigma \geq \Gamma(\delta), \delta \in \mathcal{D}(\rho) \right\} \quad (4.1)$$

$$= \inf \left\{ \text{tr } W \text{Re } \Gamma(\delta) + \text{tr} \left| \sqrt{W} \text{Im } \Gamma(\delta) \sqrt{W} \right| : \delta \in \mathcal{D}(\rho) \right\}, \quad (4.2)$$

$$\Gamma_{jk}(\delta) \equiv \text{tr } \delta_j \delta_k \rho, \quad (4.3)$$

where $|A| \equiv \sqrt{A^\dagger A}$, $\dagger$ denotes the conjugate transpose, and Re and Im denote the entry-wise real and imaginary parts, respectively. Eq. (4.2) is due to Nagaoka [1]. It can be proved from the following lemma.

Lemma 3 (Belavkin and Grishanin [9]). *Let A be any complex positive-semidefinite matrix. Then*

$$\text{tr } \text{Re } A \geq \text{tr } |\text{Im } A|. \quad (4.4)$$

Equality is achieved if $\text{Re } A = |\text{Im } A|$; $A = \text{Re } A + i \text{Im } A = |\text{Im } A| + i \text{Im } A$ remains positive-semidefinite.

Because

$$L \geq \delta \delta^\top \implies \Sigma \geq \Gamma(\delta) \quad (4.5)$$

under the constraints on L and Σ , we find [3]

$$\boxed{\text{NH}(\rho) \geq \text{HN}(\rho)}. \quad (4.6)$$

For the IID model, we know [4, Lemma 4]

$$\text{HN}(\rho^{\otimes N}) = \frac{1}{N} \text{HN}(\rho), \quad (4.7)$$

so

$$N \text{MI}_{\text{sep}}(\rho^{\otimes N}) \geq N \text{MI}(\rho^{\otimes N}) \geq \text{HN}(\rho), \quad (4.8)$$

where $\text{MI}(\rho^{\otimes N})$ is the most informative bound for the IID model and any measurement, not just separable.

5. HELSTROM BOUND

Define the Helstrom bound as

$$\text{Hel}(\rho) \equiv \inf \{ \text{tr } W \text{Re } \Gamma(\delta) : \delta \in \mathcal{D}(\rho) \} \quad (5.1)$$

$$= \text{tr } W \Sigma(\rho), \quad (5.2)$$

$$\Sigma(\rho) \equiv (\partial \beta)^\top J(\rho)^{-1} \partial \beta, \quad (5.3)$$

$$J_{jk}(\rho) \equiv \text{tr } (S_j \circ S_k) \rho, \quad A \circ B \equiv \frac{AB + BA}{2}, \quad (5.4)$$

$$\partial_j \rho(\theta) = \rho \circ S_j, \quad (5.5)$$

where $J(\rho)$ is the quantum Fisher information matrix and $\Sigma(\rho)$ is the matrix quantum CRB first proposed by Helstrom [10]. Each S_j is called a score operator, also called a symmetric logarithmic derivative (SLD)

[5]. A proof of Eq. (5.2) can be found, for example, in Ref. [11]. The $\delta \in \mathcal{D}(\rho)$ that achieves the infimum is called the efficient influence operator, given by

$$\delta_j^{\text{eff}} = \sum_{k,l} (\partial_k \beta_j) [J(\rho)^{-1}]_{kl} S_l. \quad (5.6)$$

It is known that [11, 12]

$$\boxed{2\text{Hel}(\rho) \geq \text{HN}(\rho) \geq \text{Hel}(\rho)}, \quad (5.7)$$

which can be proved from Eq. (4.2) and Lemma 3. For the IID model, it is also well known that [5]

$$J(\rho^{\otimes N}) = NJ(\rho), \quad \text{Hel}(\rho^{\otimes N}) = \frac{1}{N} \text{Hel}(\rho). \quad (5.8)$$

6. GILL–MASSAR BOUND

Theorem 3 (Gill–Massar bound [13]). *For any POVM E ,*

$$\text{tr } J(\rho)^{-1} J(\rho, E) \leq \dim \mathcal{H} - 1 \quad (6.1)$$

where $J(\rho, E)$ is the classical Fisher information matrix given by Eq. (2.11) and $J(\rho)$ is the Helstrom information matrix given by Eq. (5.4).

Before giving the proof, we need a few preliminary results. Construct the real Hilbert space $\mathcal{L}^2(\rho)$ from self-adjoint operators $\mathcal{O}(\mathcal{H})$ with the semi-inner product and the seminorm

$$\langle A, B \rangle_\rho \equiv \text{tr}(A \circ B)\rho, \quad \|A\|_\rho \equiv \sqrt{\langle A, A \rangle_\rho}, \quad (6.2)$$

respectively [7]. Since $\mathcal{H}$ is finite-dimensional, each element of $\mathcal{L}^2(\rho)$ is simply an equivalence class $A + \mathcal{N}(\rho) \equiv \{A + N : N \in \mathcal{N}(\rho)\}$ for an $A \in \mathcal{O}(\mathcal{H})$, where $\mathcal{N}(\rho)$ is the set of elements with zero seminorm $\|\cdot\|_\rho = 0$ [7]. In what follows, we will write an equivalence class $A + \mathcal{N}(\rho) \in \mathcal{L}^2(\rho)$ as A for brevity. For example, $I + \mathcal{N}(\rho)$ with the identity operator I is written as I .

Define a one-dimensional subspace $\mathcal{I}(\rho) \subseteq \mathcal{L}^2(\rho)$ as the linear span of I :

$$\mathcal{I}(\rho) \equiv \text{span}\{I\} = \{cI : c \in \mathbb{R}\} \quad (6.3)$$

and its orthocomplement relative to $\mathcal{L}^2(\rho)$ as the zero-mean subspace

$$\mathcal{Z}(\rho) \equiv \mathcal{I}(\rho)^\perp = \left\{ A \in \mathcal{L}^2(\rho) : \langle A, I \rangle_\rho = 0 \right\}. \quad (6.4)$$

In particular, since $\rho(\theta)$ is normalized, each score operator S_j defined by Eq. (5.5) is in $\mathcal{Z}(\rho)$. Define the score tangent space as the linear span of all the scores:

$$\mathcal{T}(\rho) \equiv \text{span}\{S_j\} = \left\{ \sum_j c_j S_j : c \in \mathbb{R}^p \right\} \subseteq \mathcal{Z}(\rho). \quad (6.5)$$

$\mathcal{I}(\rho)$, $\mathcal{Z}(\rho)$, and $\mathcal{T}(\rho)$ are all Hilbert spaces and subspaces of $\mathcal{L}^2(\rho)$. By basic Hilbert-space theory [14], the orthogonal projection operators onto these subspaces become

$$\Pi_{\mathcal{I}(\rho)} A = \langle A, I \rangle_\rho I, \quad (6.6)$$

$$\Pi_{\mathcal{Z}(\rho)} A = A - \Pi_{\mathcal{I}(\rho)} A = A - \langle A, I \rangle_\rho I, \quad (6.7)$$

$$\Pi_{\mathcal{T}(\rho)} A = \sum_{j,k} S_j [J(\rho)^{-1}]_{jk} \langle S_k, A \rangle_\rho, \quad (6.8)$$

where $J(\rho)$, the Helstrom information matrix, is the same as the Gram matrix of the scores $\{S_j\}$, so the invertibility of $J(\rho)$ is equivalent to the linear independence of $\{S_j\}$.

Write each element $E(\omega)$ of an arbitrary POVM E as

$$E(\omega) = \sum_s \lambda_{\omega s} |e_{\omega s}\rangle \langle e_{\omega s}|, \quad (6.9)$$

where each $|e_{\omega s}\rangle$ is an eigenvector of $E(\omega)$ with eigenvalue $\lambda_{\omega s} > 0$. We can then construct a POVM without the sum over s as

$$E'(\omega, s) = \lambda_{\omega s} |e_{\omega s}\rangle \langle e_{\omega s}|, \quad (6.10)$$

where each element $E'(\omega, s)$ of E' is rank-1. By monotonicity, $J(\rho, E) \leq J(\rho, E')$, so any upper bound on $J(\rho, E')$ for rank-1 POVMs is an upper bound for any POVM.

Proof of Theorem 3. Write the left-hand side of Eq. (6.1) as

$$\text{tr } J(\rho)^{-1} J(\rho, E) = \sum_{\omega, j, k} [J(\rho)^{-1}]_{jk} \frac{[\text{tr } E(\omega) \partial_j \rho(\theta)] [\text{tr } E(\omega) \partial_k \rho(\theta)]}{\text{tr } E(\omega) \rho} \quad (6.11)$$

$$= \sum_{\omega, j, k} [J(\rho)^{-1}]_{jk} \frac{\langle E(\omega), S_j \rangle_\rho \langle S_k, E(\omega) \rangle_\rho}{\langle E(\omega), I \rangle_\rho} = \sum_{\omega} \frac{\langle E(\omega), \Pi_{\mathcal{T}(\rho)} E(\omega) \rangle_\rho}{\langle E(\omega), I \rangle_\rho} \quad (6.12)$$

$$\leq \sum_{\omega} \frac{\langle E(\omega), \Pi_{\mathcal{Z}(\rho)} E(\omega) \rangle_\rho}{\langle E(\omega), I \rangle_\rho} = \sum_{\omega} \frac{\langle E(\omega), E(\omega) - \langle E(\omega), I \rangle_\rho I \rangle_\rho}{\langle E(\omega), I \rangle_\rho} \quad (6.13)$$

$$= \left[\sum_{\omega} \frac{\text{tr } E(\omega)^2 \rho}{\text{tr } E(\omega) \rho} \right] - 1. \quad (6.14)$$

The inequality above comes from the fact that any projection operator Π is self-adjoint and idempotent ($\Pi^2 = \Pi$), so

$$\langle E(\omega), \Pi_{\mathcal{T}(\rho)} E(\omega) \rangle_\rho = \|\Pi_{\mathcal{T}(\rho)} E(\omega)\|_\rho^2 \leq \|\Pi_{\mathcal{Z}(\rho)} E(\omega)\|_\rho^2 = \langle E(\omega), \Pi_{\mathcal{Z}(\rho)} E(\omega) \rangle_\rho, \quad (6.15)$$

where the last inequality is due to the Pythagorean theorem. For any rank-1 POVM,

$$E(\omega)^2 = [\text{tr } E(\omega)] E(\omega), \quad (6.16)$$

$$\sum_{\omega} \frac{\text{tr } E(\omega)^2 \rho}{\text{tr } E(\omega) \rho} = \sum_{\omega} \text{tr } E(\omega) = \text{tr } I = \dim \mathcal{H}. \quad (6.17)$$

Plugging this result into Eq. (6.14) proves the theorem. $\square$

Lemma 1 allows us to apply the bound to the IID model with separable measurements as well:

$$\text{tr } [J(\rho)^{-1} J(\rho^{\otimes N}, E)] \leq N(\dim \mathcal{H} - 1) \quad \forall \text{ separable } E. \quad (6.18)$$

An alternative form of the Gill–Massar bound is [13]

$$\text{NMI}_{\text{sep}}(\rho^{\otimes N}) = \text{MI}(\rho) \geq \frac{1}{\dim \mathcal{H} - 1} \left[\text{tr } \sqrt{J(\rho)^{-1/2} (\partial \beta) W (\partial \beta)^\top J(\rho)^{-1/2}} \right]^2. \quad (6.19)$$

By monotonicity, $J(\rho, E) \leq J(\rho)$ for any POVM E , so

$$\text{tr } J(\rho)^{-1} J(\rho, E) \leq \text{tr } J(\rho)^{-1} J(\rho) = p, \quad (6.20)$$

where p is the dimension of the parameter space Θ . The Gill–Massar bound is a tighter bound only when $p > \dim \mathcal{H} - 1$.

There does not seem to be any other general relationship between the Gill–Massar bound and the other bounds.

Unfortunately, the Gill–Massar bound is useless when $\dim \mathcal{H} = \infty$. The Nagaoka–Hayashi bound, on the other hand, seems generalizable to the infinite-dimensional case, although numerics is still required to compute the bound.

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